Supplementary File 4

Phase 4b inter-phase reconciliation log

Five inter-phase discrepancies were flagged during the structured reconciliation pass conducted at the end of the Phase 4b quality-appraisal phase. Each was resolved by direct re-verification against the source PDF before figures and tables were generated. This appendix records every discrepancy and its resolution.

Item 1: Paper 8 (PMID 38061942, gallbladder, Meng 2024)

Issue: Phase 4a originally coded External_Validation = NO. Phase 4b PROBAST appraisal flagged ambiguity around train/test center separation.

Resolution: PDF Methods + Figure 2 confirmed center-based train/test split (Center 2 trained n=105, Center 1 tested n=62). Phase 4a row 8 updated to External_Validation = YES, Cohort_External_Test = 62, Cohort_Internal_Test = NR. RQS Item 12 stayed at +3 (already correct). Corpus-wide external-validation count: 10/28 -> 11/28. Author caveat: paper's own Discussion calls this 'lacked external test cohorts' (treating sister-center within Shanxi Medical University network as internal). Strict PROBAST/RQS interpretation = external single-site; flagged for sensitivity analysis.

Item 2: Paper 10 (PMID 40386715, lung HGP, Chen 2025)

Issue: Cohort columns appeared inconsistent: corpus showed Cohort_Train=311 but the paper described 213/92/98 sub-cohorts at a single institution.

Resolution: PDF Methods confirmed single-center (Taizhou Hospital of Zhejiang Province) with three temporal sub-cohorts: 213 = train (Jan-Oct 2023), 92 = internal validation/tuning (Jan-Oct 2023), 98 = internal test (temporal hold-out Nov 2023 - Jan 2024). Two CT scanners (GE Discover CT750 HD + UCT 530) at one site = scanner diversity, not multi-center. Phase 4a row 10 updated to Cohort_Train=213, Cohort_Internal_Test=98, Cohort_External_Test ='-', External_Validation=NO. The 92 internal-validation cohort was folded into Notes since the CSV schema lacks a Cohort_Internal_Val column.

Item 3: Paper 14 (PMID 37495426, osteoporosis BLS, Zhang 2024)

Issue: Methods describes scanner-vendor segregation between train (Philips DigitalDiagnost DR) and test (Siemens AXIOM AristosVX DR), but does not explicitly name which of the three centers contribute to training versus test. Phase 4a coded External_Validation=NO; Phase 4b RQS Item 12 was tentatively coded +3.

Resolution: PROBAST and TRIPOD require explicit institutional separation for cross-vendor scanner data to count as external validation. Strict reading applied: External_Validation stays NO; RQS Item 12 downgraded from +3 to +2 (internal). RQS total revises from 6 to 5 (new corpus low). Methods also confirmed NO feature reduction (no LASSO/Pearson/mRMR/ICC) and NO multiple-testing correction; the BLS random-projection layer expands ~85 features to 872 derived nodes - not a valid RQS Item 5 technique. Item 5 = -3 confirmed correct.

Item 4: Paper 24 (PMID 32941954, ESCC, Hu 2021)

Issue: Paper reports two radiomics AUCs (0.725 and 0.712). Phase 4a Rad_Arm_Value captured 0.725; Phase 4b CLAIM cross-check raised the question of which arm is the primary radiomics comparator.

Resolution: PDF Methods (Section 'ROIs') confirmed 0.725 is the PRIMARY radiomics arm (3-slice ROI matched to DL input geometry). 0.712 is the SECONDARY radiomics arm (whole-volume ROI, described as 'comparison' in Methods). Phase 4a row 24 Rad_Arm_Value=0.725 was already correct. No data changes; Notes updated to capture the primary/secondary distinction so the 0.712 is not mis-pulled in future analyses. Delta = +0.080 confirmed.

Item 5: Paper 26 (PMID 40371075, stroke MRI, Pei 2025)

Issue: Phase 4a FM_Pretraining recorded as 'NR' (not reported). Phase 4b CLAIM appraisal Item 22 (training data provenance) required a definitive coding.

Resolution: PDF Methods Section 2.4 confirmed transfer learning but does not name the pretraining corpus anywhere. Two defensible interpretations: (a) strict NR with Limitations flag, or (b) pragmatic inference of ImageNet default (PyTorch + ResNet101 + transfer learning with no cited corpus). Option (b) applied per scoring recommendation. FM_Pretraining updated from 'NR' to 'ImageNet (inferred - not explicitly stated)'. Tier classification = ImageNet-CNN (unchanged from working assumption). Limitations flag added: 'Pretraining corpus not reported despite explicit transfer learning; PyTorch ImageNet default presumed but not coded.'

Summary of post-reconciliation corpus state

After all five items were resolved, the corpus state for Phase 5 statistical synthesis was: 28/28 papers scored on RQS, CLAIM and PROBAST; external-validation count = 11/28 (39%); RQS range 5 to 20, mean 12.07, median 12.0; PROBAST overall = 25 High / 3 Unclear / 0 Low.